Initialize controller: `fcs_init()`
Zero controller state; set mode to DISARMED; clear armed/airborne/HOLD flags; initialize RC history to 1500 µs, except throttle, HOLD and flaps, which are initialized to 1000 µs.

Main loop
Wait for next 100 Hz tick; read RC frame, air data (pitot, baro/AGL) and current time. In `fcs_run()`, use $\Delta t = 0.01$ s.

1. Validate the RC frame
Frame age: unsigned (`now_ms` $-$ `stamp_ms`) $<$ 250 ms.
Every channel pulse must be between 880 and 2120 µs.

Valid frame
Copy all channels to `rc_last`; set `rc_bad`$=0$; increment `rc_good`.

Invalid frame
Keep last good channel values; set `rc_good`$=0$; increment `rc_bad`.

Debounce link status
If `rc_ok` and `rc_bad` $\geq$ 10: set `rc_ok` = false.
Else if not `rc_ok` and `rc_good` $\geq$ 30: set `rc_ok` = true.
Otherwise retain the current link status.

2. Validate and filter air data
Sample must be fresh ($<$ 200 ms), flagged valid and finite. Airspeed: $0 \leq V < 60$ m/s. Altitude: $-100 < h < 10000$ m.
Airspeed: low-pass, $\tau = 0.20$ s.
Altitude: reject jumps $>$ 10 m (accepted if they persist 50 samples); low-pass, $\tau = 0.25$ s. Vertical speed $v_z = \mathrm{d}h/\mathrm{d}t$: low-pass, $\tau = 0.60$ s, clamped to ±15 m/s.

Sensor faults
Airspeed stale or implausible: speed protections off, gains nominal, $V = V_{\mathrm{ref}}$ for scheduling. Altitude stale or implausible: altitude and flight-path protections off. Filters re-initialise when a sensor returns.

3. "Flying" latch from airspeed (frozen if airspeed invalid)
Set `airborne` when $V > 1.2\,V_{s,\mathrm{clean}}$.
Clear `airborne` and `armed` if $V < 0.5\,V_{s,\mathrm{clean}}$ for more than 2 s.

4. Decode pilot inputs from last good frame
CH1 aileron, CH2 elevator, CH3 throttle, CH4 rudder, CH5 HOLD, CH6 flaps.
Aileron/elevator/rudder: normalize to $[-1,1]$, apply deadband and expo; apply elevator sign. Throttle: normalize to $[0,1]$.
HOLD switch is high if CH5 $>$ 1700 µs. Flaps: CH6 $<$ 1300: up; 1300–1699: half; $\geq$ 1700: full.

5. Arm interlock
If airborne: set `armed` = true.
Else if not armed, link good and throttle $<$ 0.05 (idle once after power-up): set `armed` = true.

6. Evaluate HOLD switch and latch state
If link bad, not armed, not airborne, or HOLD switch low: clear HOLD; if the switch is low, also clear `hold_lockout`. Otherwise, on a rising edge (previous switch low, current high) and not locked out: engage HOLD and latch shaped aileron, elevator, rudder, throttle stick and flap selector.

Clear HOLD; set `hold_lockout` = true. Toggle HOLD switch low to unlock.

Keep the current HOLD state.

Update `hold_sw_prev`; then select the mode (step 7) and the commands for that mode (step 8).

Key configuration values
RC timeout: 250 ms; air-data timeout: 200 ms
Link loss: 10 bad ticks; recovery: 30 good ticks
Arm threshold: throttle < 0.05
HOLD breakout: 0.8 aileron/elevator deflection
V_s clean / full flaps: 11 / 9.5 m/s
V_{fe} full / half flaps: 16 / 19 m/s
V_{ne}: 30 m/s; V_{ref}: 18 m/s
Ceiling: 120 m (20 m margin); floor: disabled (0)
Sink limit: 4 m/s; flight-path limits: ±15°
Bank limit: ±45°; pitch: $+20^\circ/-15^\circ$
Aileron–rudder mix: 0.25; flap–elevator: -0.05
Servo travel: 500 µs from centre

FAILSAFE
Airborne: aileron = `fs_ail` = 0.10; elevator = clamp$(0.04\,(V-15), \pm 0.3)$, or `fs_ele` = 0 without airspeed; rudder = 0; throttle = `fs_thr` = 0.35; flaps = 0.
On the ground: all outputs 0.

DISARMED
Shaped sticks for aileron/elevator/rudder.
Throttle forced to zero.
Flaps follow the selector.

GROUND
Armed, not yet flying.
Direct control: shaped sticks, throttle and flap selector.
No protections.

HOLD
Use latched aileron, elevator, rudder and flaps.
Throttle = min(latched throttle, current throttle stick).

NORMAL
Shaped sticks and throttle.
Flaps follow the selector.
Full protection set applies.

Protection scope
`prot` = airborne and mode is not DISARMED or GROUND (so FAILSAFE, HOLD and NORMAL are protected).
`use_v` = prot and airspeed OK; `use_h` = prot and altitude OK.

9. Airspeed protection terms (only if `use_v`)
$V_{s,\mathrm{eff}} = \mathrm{lerp}(V_{s,\mathrm{clean}}, V_{s,\mathrm{full}}, \mathit{flap}) \big/ \sqrt{\max(\cos\phi_{\mathrm{est}}, 0.5)}$.
Gain schedule: $\mathit{sched} = \mathrm{clamp}\big((V_{\mathrm{ref}}/\max(V, V_{\min}))^2, 0.5, 1.8\big)$.
Stall: k_{stall} ramps $0 \to 1$ from $1.15\,V_{s,\mathrm{eff}}$ to $1.30\,V_{s,\mathrm{eff}}$; nose-down bias $b_{\mathrm{stall}} = -0.25 \cdot \mathrm{clamp}\big((1.15V_{s,\mathrm{eff}} - V)/(0.10V_{s,\mathrm{eff}}), 0, 1\big)$.
Overspeed: k_o ramps $0 \to 1$ from $0.90\,V_{ne}$ to V_{ne}; nose-down authority $\times(1 - k_o)$; pull-up bias $b_{\mathrm{over}} = 0.10\,k_o$.

10. Altitude and flight-path terms (only if `use_h`)
Ceiling: k_c ramps $0 \to 1$ over the 20 m below 120 m; up-elevator $\times(1-k_c)$; nose-down bias $b_{\mathrm{ceil}} = -0.10 \cdot \mathrm{clamp}\big((h - h_{\mathrm{ceil}})/20\,\mathrm{m}, 0, 1\big)$.
Floor (if $h_{\mathrm{floor}} > 0$, $h < h_{\mathrm{floor}}$ and not landing, i.e. flaps < 0.45 and throttle > 0.20): down-elevator fades out; if sink > 4 m/s add pull-up bias $0.20\,k$ and throttle floor $0.7\,k$, with $k = \mathrm{clamp}\big((-v_z - 4)/2, 0, 1\big)$.
Flight-path angle (if `use_v` too): $\gamma = \arcsin\big(v_z/\max(V, V_{\min})\big)$.

11. Shape surface commands
$\mathit{ail} \mathrel{\times=} \mathit{sched} \cdot \mathrm{lerp}(0.5, 1, k_{\mathrm{stall}})$; $\mathit{ele} \mathrel{\times=} \mathit{sched}$; $\mathit{rud} \mathrel{\times=} \mathit{sched}$.
Aileron limit interpolates from 0.80 (flaps up) to 0.55 (full flaps), based on the previous output flap position.
Elevator clamped to $[-0.55, +0.70]$; rudder to $[-0.85, +0.85]$.

Reset `bank_est` and `pitch_est` to zero.

Apply protections
Up-elevator $\times\, k_{\mathrm{stall}} k_{\mathrm{ceil}}$; down-elevator $\times\,(1 - k_o)\,k_{\mathrm{floor\_dn}}$.
Taper aileron near estimated bank limits of ±45°; taper elevator near estimated pitch limits of $+20^\circ$ and -15°; if `use_h` and `use_v`, also taper elevator near measured γ limits of ±15°.
Add biases: $\mathit{ele} \mathrel{+=} b_{\mathrm{stall}} + b_{\mathrm{over}} + b_{\mathrm{ceil}} + b_{\mathrm{floor}}$; clamp to $[-1, 1]$.
Integrate assumed rates and self-levelling:
$$\dot b \approx k_b\,\mathrm{ail} - b/\tau_b, \qquad \dot p \approx k_p\,\mathrm{ele} - p/\tau_p$$

Remaining open-loop part
Bank and pitch limiters still use dead-reckoned estimates that drift with wind, trim and CG.
The flight-path limiter uses the measured γ.

12. Throttle (protected modes)
Ceiling cap: $\mathit{thr} \leftarrow \min(\mathit{thr}, \mathrm{lerp}(1, 0.40, k_c))$.
Low-speed power floor (if `use_v`): $\mathit{thr} \leftarrow \max(\mathit{thr}, 0.85\,k_t)$, $k_t = \mathrm{clamp}\big((1.25V_{s,\mathrm{eff}} - V)/(0.25V_{s,\mathrm{eff}}), 0, 1\big)$.
Sink-rate floor: $\max(\mathit{thr}$, `thr_floor`$)$; overspeed cut: $\mathit{thr} \mathrel{\times=} (1 - k_o)$.
Output = $\mathit{thr} \times \mathit{thr\_max}$ (1.0) if armed, otherwise 0.

13. Flap speed limits (if `use_v` and not FAILSAFE)
Allowed flaps: 0 if $V > 19$ m/s; 0.5 if $V > 16$ m/s; otherwise 1.
Flap command is limited to the allowed value. Do not retract flaps when $V < 1.3\,V_{s,\mathrm{clean}}$, unless the speed limit forces it.

14a. Control mixing and clamp
Target aileron = aileron command.
Target elevator = elevator $-\,0.05\times$ previous flap output.
Target rudder = rudder $+\,0.25\times$ aileron.
Target throttle = clamp$(\mathit{thr}, 0, 1)$; target flap = clamp$(\mathit{flap}, 0, 1)$.
Aileron/elevator/rudder targets clamped to $[-1, +1]$.

14b. Slew-limit outputs
Aileron/elevator/rudder slew: 8 units/s.
Throttle rise: 2 units/s; throttle fall: 3 units/s.
Flap slew: 0.4 units/s.

14c. Apply servo reversals
Reverse aileron, elevator and rudder output if the corresponding configuration flag is enabled.

14d. Convert normalized outputs to RC pulses
Aileron/elevator/rudder: 1500 + output × 500 µs.
Throttle/flaps: 1000 + output × 1000 µs.
CH5 status: 2000 µs if HOLD active; otherwise 1000 µs.